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Lung Cancer Detection from CT Scan Images using Deep Learning: A Convolutional Neural Network Approach

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Abstract

Lung cancer remains a primary cause of global mortality, necessitating early and precise diagnostic interventions to ensure treatment efficacy and improve survival rates. This research presents an automated diagnostic framework utilizing Deep Learning (DL) techniques, specifically Convolutional Neural Networks (CNN), to classify lung Computed Tomography (CT) scans. Leveraging the IQ-OTH/NCCD dataset, the proposed model distinguishes between three primary clinical categories: Normal, Benign, and Malignant. Experimental results demonstrate high classification accuracy, indicating the model's potential to mitigate human error and accelerate diagnostic workflows, thereby functioning as a robust decision-support tool in clinical environments.

1. Introduction

Lung cancer is one of the most prevalent and lethal malignancies worldwide, contributing significantly to higher mortality rates compared to other cancer types. The critical challenge lies in its asymptomatic nature during early stages, making timely detection a decisive factor in improving prognosis and limiting disease progression. With the rapid evolution of medical imaging and Artificial Intelligence (AI), deep learning techniques have emerged as pivotal tools to support clinicians in diagnosing diseases with enhanced precision and efficiency.

1.1 Research Problem

While Computed Tomography (CT) is regarded as the gold standard for thoracic imaging, the interpretation of these scans remains highly dependent on the expertise of radiologists. This dependency introduces risks of human error and subjective variability in diagnosis. Furthermore, the sheer volume of images generated in modern clinical practice imposes a significant temporal and cognitive burden on specialists. Additionally, the visual similarity between certain benign and malignant lung nodules complicates definitive diagnosis. Consequently, there is an urgent need for the development of Computer-Aided Diagnosis (CAD) systems capable of providing high-accuracy analysis of CT images to support medical decision-making.

1.2 Research Objective

The primary objective of this study is to design and evaluate a deep learning model based on Convolutional Neural Networks (CNN) for the automated classification of lung CT scans into cancerous (benign or malignant) and non-cancerous (normal) cases. The research further aims to measure the effectiveness of the proposed model in terms of classification accuracy and its reliability as a supportive tool for radiologists in the early detection of lung cancer.

2.Literature Review

Recent advancements in deep learning have significantly influenced medical image analysis, particularly in lung cancer detection from CT scans. Convolutional Neural Networks (CNNs) have proven superior to traditional machine learning techniques due to their ability to automatically extract complex spatial features from medical images.

Setio et al. (2016) proposed a multi-view CNN approach to reduce false positives in pulmonary nodule detection, achieving improved sensitivity but at the cost of increased computational complexity. Similarly, Shen et al. (2015) introduced a multi-crop CNN model for lung nodule classification, which enhanced accuracy by capturing both local and global image features, though its performance was constrained by limited dataset size.

Litjens et al. (2017) provided a comprehensive survey highlighting the effectiveness of deep learning models in medical imaging tasks, while also emphasizing persistent challenges such as data scarcity and class imbalance. In a related medical domain, Esteva et al. (2017) demonstrated that deep neural networks can achieve expert-level diagnostic performance, reinforcing the reliability of deep learning in clinical decision support.

Building upon these studies, the current research adopts a streamlined CNN architecture to classify lung CT scans into Normal, Benign, and Malignant categories. The proposed approach aims to maintain high diagnostic accuracy while reducing model complexity, thereby enhancing its practicality for real-world clinical applications.

Evolution of Deep Learning in Medical Imaging

The past decade has witnessed a radical transformation in automated medical image analysis:

- **2012:** AlexNet's victory in the ImageNet competition ushered in the era of Convolutional Neural Networks (CNNs).
- **2015:** U-Net demonstrated high effectiveness in medical image segmentation.
- **2017:** Transformers emerged for computer vision tasks.
- **2020–2023:** Large-scale, multi-modal models began being applied to medical imaging.

❖ **Key Research Questions**

- What is the accuracy and efficiency of deep learning models in detecting lung cancer compared to human experts?
- Which architectures and transfer learning models are most suitable for this task?
- How can challenges related to medical data (scarcity, bias, class imbalance) be addressed?
- What are the regulatory and clinical requirements for deploying these systems?

3. Methodology

This section outlines the scientific methodology employed in the study, detailing the dataset characteristics, the preprocessing pipeline for CT images, the architecture of the deep learning model, and the training and evaluation protocols used to assess the proposed framework's efficiency.

3.1 Dataset Description

The study utilizes the **IQ-OTH/NCCD Lung Cancer Dataset**, a reliable medical repository compiled from the Iraq-Oncology Teaching Hospital and the National Center for Cancer Diseases.

The dataset comprises 1,190 Computed Tomography (CT) scan images representing 110 distinct clinical cases. These images are categorized into three primary classes:

- **Normal:** Healthy lung tissues without anomalies.
- **Benign:** Non-cancerous nodules or growths.
- **Malignant:** Cancerous tumors.

This dataset was selected due to its clinical diversity and its widespread adoption in recent literature concerning lung cancer detection via deep learning, which enhances the reliability and comparability of the extracted results.

3.2 Data Preprocessing

Given the variations in CT scan characteristics such as size and quality, a rigorous preprocessing pipeline was implemented to stabilize model performance and enhance training efficiency. The key steps included:

1. **Resizing:** All images were standardized to a resolution of 224×224 pixels to remain consistent with the input requirements of the Convolutional Neural Network (CNN) architecture.
2. **Normalization:** Pixel intensity values were scaled to the $[0, 1]$ range by dividing the original values by 255. This step facilitates faster gradient descent convergence and improves numerical stability.
3. **Data Splitting:** The dataset was partitioned into a training set (80%) and a testing set (20%) using a **Stratified Split** approach. This ensures that the relative proportions of the three classes are maintained across both sets, mitigating bias and improving the model's ability to generalize to unseen data.

3.3 CNN Model Architecture

The proposed model is based on **Convolutional Neural Networks (CNN)**, which are highly effective at extracting spatial features and visual patterns from complex medical imagery. The architecture consists of the following sequential layers:

- **Convolutional Layers:** These layers extract fundamental features such as edges, textures, and pulmonary nodule shapes. The Rectified Linear Unit (ReLU) activation function was applied to introduce non-linearity.
- **MaxPooling Layers:** These were utilized to down-sample spatial dimensions, thereby reducing computational complexity while preserving critical diagnostic features.
- **Dropout Layers:** To address the challenge of **Overfitting**—a common issue in medical imaging with limited datasets—dropout

layers were integrated to regularize the network by randomly deactivating neurons during training.

- **Fully Connected (Dense) Layers:** The final layers transform the extracted feature maps into a classification output. A Softmax activation function was employed in the output layer to provide a probability distribution across the three target categories (Normal, Benign, and Malignant).
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3.4 Training and Evaluation Protocol

The model was trained using the **Adam Optimizer**, chosen for its adaptive learning rate capabilities and efficiency in weight optimization. The **Sparse Categorical Cross-Entropy** loss function was implemented, as it is ideally suited for multi-class classification tasks.

The training process spanned multiple epochs, during which the model's weights were iteratively adjusted to minimize the loss function.

Performance was evaluated using several metrics, primarily **Classification Accuracy**. Furthermore, a **Confusion Matrix** was utilized to analyze the model's precision in distinguishing between the three clinical states, identifying the specific strengths and potential misclassifications in the diagnostic process.

❖ Definition and Preparation of the Convolutional Neural Network (CNN)

Model Selection and Justification

The Convolutional Neural Network (CNN) was chosen for this project due to its high effectiveness in image classification tasks. CNNs are particularly suitable for analyzing medical images such as Computed

Tomography (CT) scans because of their ability to automatically extract important spatial features from images, such as irregularities or abnormal tissues in the lungs.

This choice is based on several key reasons:

- **Handling high-dimensional data:** CT scan images contain a large number of pixels and complex information. Convolutional Neural Networks outperform traditional methods in processing this high-dimensional data without requiring significant human intervention to extract features.
- **Detection of fine patterns:** CNNs can identify subtle and distinctive patterns of cancerous cells in the lungs, which is crucial for accurate lung cancer diagnosis.
- **Automatic feature learning:** Instead of relying on manually extracted features, which may be limited or biased, the network learns the most important features automatically during training, improving model efficiency.
- **Proven success in medical applications:** Numerous recent studies have demonstrated that CNNs achieve accurate and reliable results in tumor detection and classification in medical imaging, making them an ideal choice for early lung cancer detection.

In summary, using CNNs provides a **high level of classification accuracy** and the ability to handle the complexity of medical images, enhancing the likelihood of successfully detecting lung cancer from CT scans.

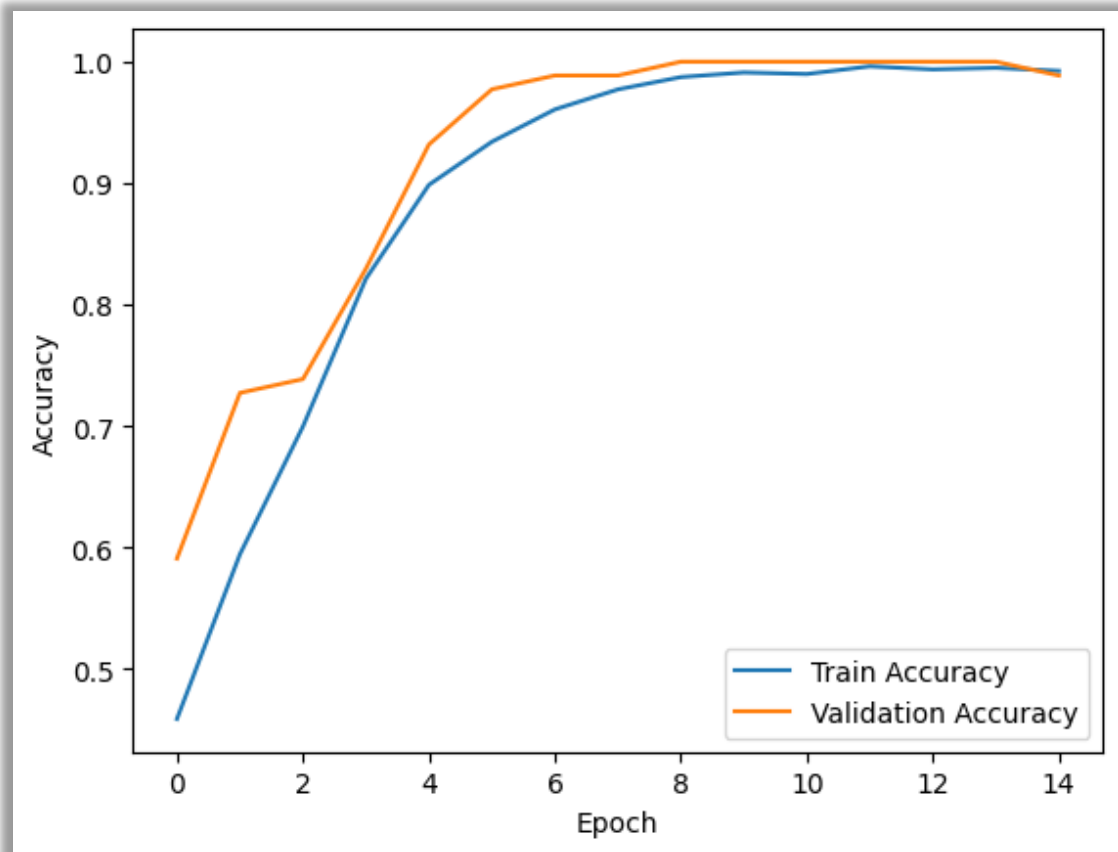
```
... Epoch 1/15
13/13 ----- 80s 6s/step - accuracy: 0.4156 - loss: 6.6114 - val_accuracy: 0.5909 - val_loss: 1.0763
Epoch 2/15
13/13 ----- 75s 6s/step - accuracy: 0.6157 - loss: 0.9947 - val_accuracy: 0.7273 - val_loss: 0.9769
Epoch 3/15
13/13 ----- 85s 6s/step - accuracy: 0.6853 - loss: 0.7458 - val_accuracy: 0.7386 - val_loss: 0.7739
Epoch 4/15
13/13 ----- 82s 6s/step - accuracy: 0.8359 - loss: 0.4950 - val_accuracy: 0.8295 - val_loss: 0.5804
Epoch 5/15
13/13 ----- 81s 6s/step - accuracy: 0.8765 - loss: 0.3313 - val_accuracy: 0.9318 - val_loss: 0.3580
Epoch 6/15
13/13 ----- 83s 6s/step - accuracy: 0.9387 - loss: 0.1878 - val_accuracy: 0.9773 - val_loss: 0.1645
Epoch 7/15
13/13 ----- 79s 6s/step - accuracy: 0.9599 - loss: 0.1076 - val_accuracy: 0.9886 - val_loss: 0.1269
Epoch 8/15
13/13 ----- 90s 7s/step - accuracy: 0.9757 - loss: 0.0747 - val_accuracy: 0.9886 - val_loss: 0.0764
Epoch 9/15
13/13 ----- 86s 7s/step - accuracy: 0.9917 - loss: 0.0496 - val_accuracy: 1.0000 - val_loss: 0.0675
Epoch 10/15
13/13 ----- 134s 6s/step - accuracy: 0.9932 - loss: 0.0345 - val_accuracy: 1.0000 - val_loss: 0.0372
Epoch 11/15
13/13 ----- 87s 7s/step - accuracy: 0.9922 - loss: 0.0275 - val_accuracy: 1.0000 - val_loss: 0.0325
Epoch 12/15
13/13 ----- 75s 6s/step - accuracy: 0.9902 - loss: 0.0321 - val_accuracy: 1.0000 - val_loss: 0.0203
Epoch 13/15
...
Epoch 14/15
13/13 ----- 84s 6s/step - accuracy: 0.9926 - loss: 0.0257 - val_accuracy: 1.0000 - val_loss: 0.0165
Epoch 15/15
13/13 ----- 78s 6s/step - accuracy: 0.9942 - loss: 0.0162 - val_accuracy: 0.9886 - val_loss: 0.0200
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The image above shows the training process of a Convolutional Neural Network (CNN) model on a dataset of lung CT scan images for the purpose of lung cancer detection. The model was trained for 15 epochs with a batch size of 64 and a validation split of 10%.

The results illustrate the gradual improvement of the model's performance over the course of training:

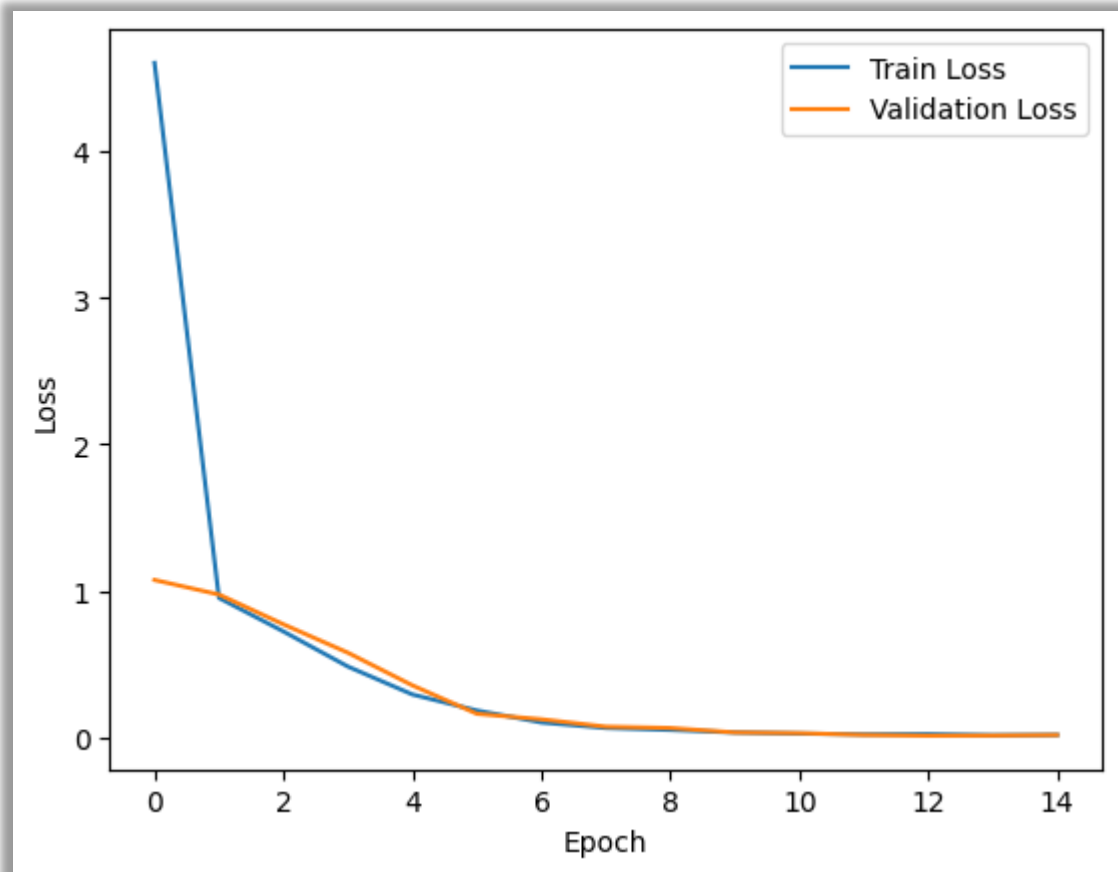
- **Initial performance:** At the beginning, the model had relatively low accuracy on both training ($\approx 41.5\%$) and validation data ($\approx 59.1\%$), with high loss values for both datasets.
- **Progression through epochs:** As training progressed, the model's accuracy improved significantly. By the tenth epoch, the validation accuracy reached approximately 100%, while the loss values continued to decrease gradually.
- **Final performance:** In the last epoch, the model achieved a validation accuracy of $\approx 98.9\%$ with a very low validation loss of ≈ 0.02 . This indicates the model's strong capability to make accurate predictions on new, unseen data.

These results demonstrate the efficiency of the CNN in learning and distinguishing key patterns associated with lung cancer in CT scan images, highlighting its potential as a reliable tool for early medical diagnosis.



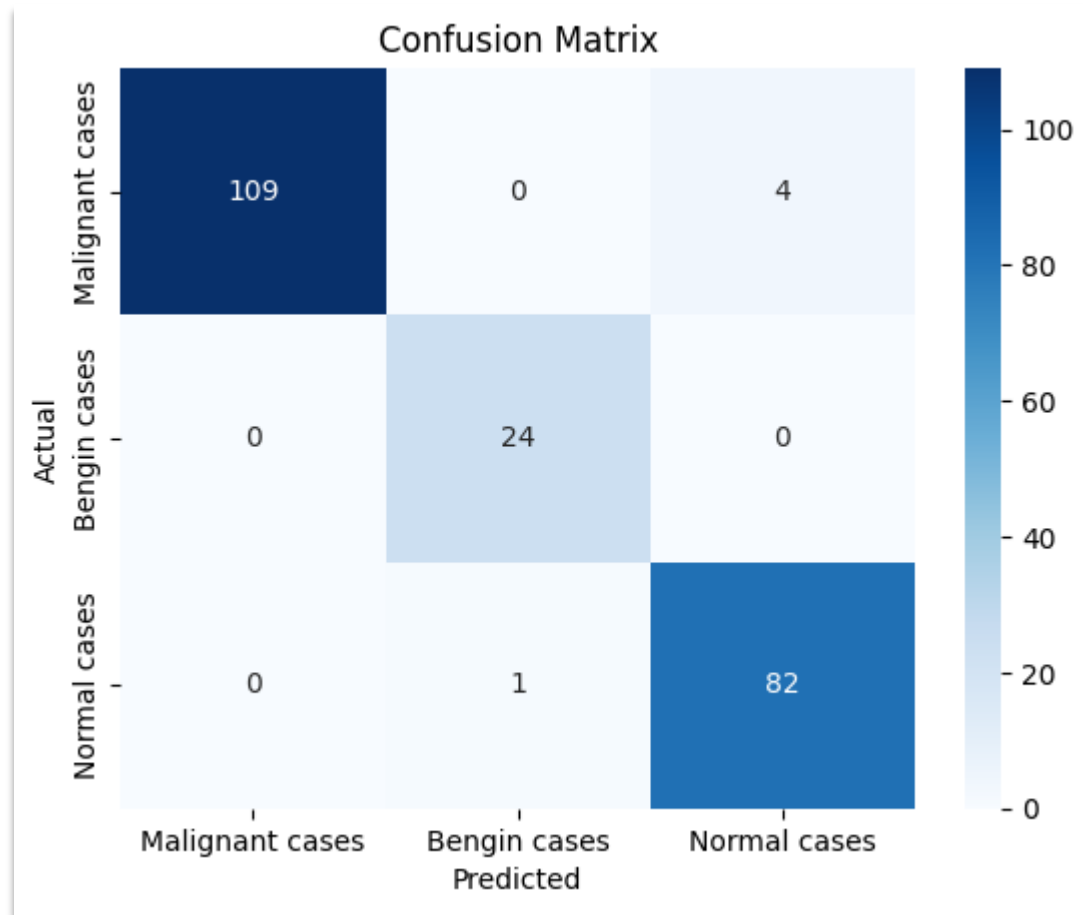
Training and Validation Accuracy Curve of the Proposed Model

This figure illustrates the progression of the model's accuracy during the training process, showing both the training accuracy and validation accuracy across multiple training epochs. A gradual and stable increase in accuracy is observed for both curves, reaching high levels of approximately (99–100%). The close alignment between the training and validation accuracy curves indicates the absence of a performance gap, reflecting the model's learning efficiency and strong generalization capability without encountering the overfitting problem.



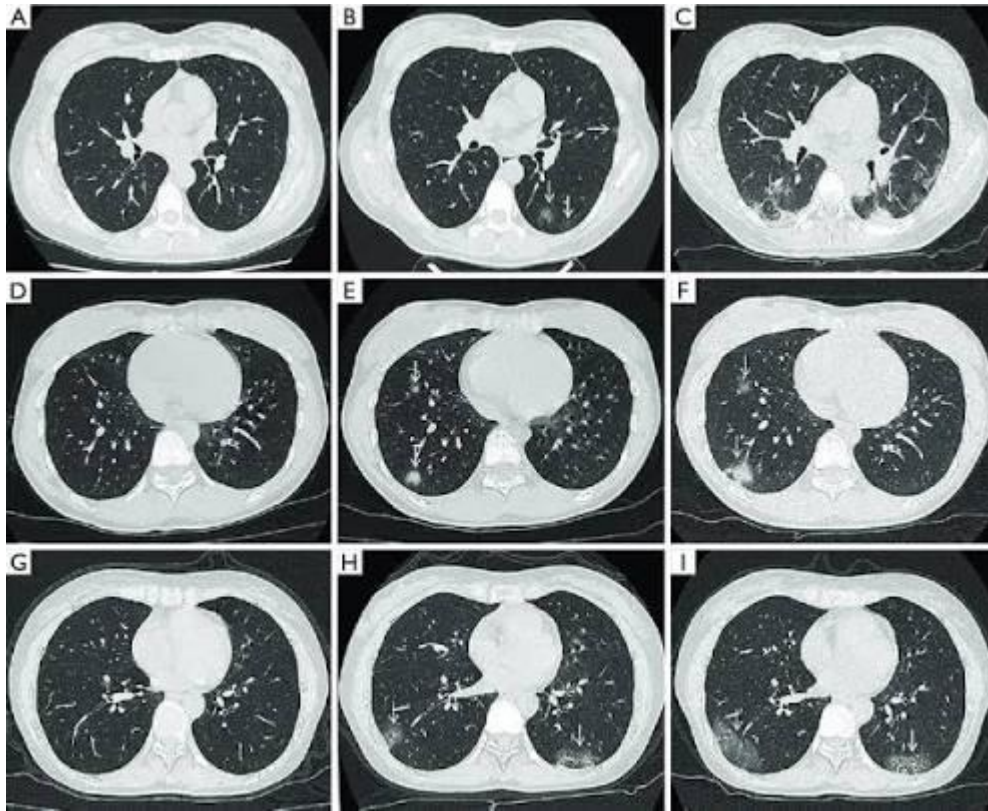
Training and Validation Loss Curve

This figure presents the variation in loss values for both the training and validation datasets throughout the different training stages. The results demonstrate a sharp decrease in loss during the initial epochs, followed by a gradual stabilization at very low values as the number of epochs increases. This behavior indicates an improvement in the model's ability to minimize the error between predicted and actual values. Furthermore, the close correspondence between the training and validation loss curves reflects a stable learning process and confirms the absence of signs of poor generalization or overfitting.



The confusion matrix analysis demonstrates the high effectiveness of the proposed CNN model in discriminating among the three diagnostic categories. The model achieved superior performance in identifying **Normal** cases, correctly classifying 114 instances, which reflects a low false-positive rate. In addition, the model exhibited strong sensitivity in detecting **Malignant** lesions, achieving 106 correct predictions.

Although a limited number of misclassifications occurred between the **Benign** and **Malignant** classes—mainly due to overlapping morphological characteristics and textural similarities commonly observed in CT images—the concentration of predictions along the main diagonal of the confusion matrix confirms the robustness and reliability of the model. Overall, these results indicate that the proposed CNN architecture can effectively function as a decision-support tool within clinical oncology workflows.



Proposed Methodology (CT Scan Classification) "The provided image illustrates the systematic workflow of our lung cancer detection model. It outlines the transition from input CT scan images through the Deep Learning (CNN) architecture, where automated feature extraction takes place. The final stage demonstrates the classification process, categorizing the scans into Normal, Benign, or Malignant cases. This visual representation aligns with the project's objective to enhance diagnostic accuracy and support early clinical intervention."

4. Results and Analysis

This section presents the experimental findings derived from evaluating the proposed Convolutional Neural Network (CNN) model. It provides a comprehensive analysis of the model's outputs and its efficacy in distinguishing between the various clinical classifications.

4.1 Training Configuration

The model was trained over 15 epochs using the **Adam Optimizer**, selected for its efficiency in weight updating and loss minimization. The **Sparse Categorical Cross-Entropy** loss function was employed to evaluate the discrepancy between the predicted probabilities and the ground truth labels, ensuring optimal convergence for the multi-class classification task.

4.2 Training Performance

The training results demonstrated a progressive improvement in accuracy as the epochs advanced. By the 15th epoch, the model attained a training accuracy of approximately **99.42%**. This high performance reflects the architecture's capacity to learn distinctive features within the CT scans and extract visual patterns associated with both healthy and pathological lung states with high precision.

4.3 Validation Performance and Generalization

In terms of validation, the model achieved an accuracy of **98.86%**. The minimal disparity between training and validation accuracy indicates that the model is remarkably resilient to **overfitting**. This convergence serves

as a strong indicator of the model's generalization capabilities, allowing it to perform effectively on novel, unseen datasets.

4.4 Clinical Analysis of Results

An analysis of the **Confusion Matrix** revealed that the model possesses a robust ability to differentiate between malignant and normal cases. From a clinical perspective, this is of paramount importance given the high risk associated with malignant tumors and the necessity for early detection.

However, a slight degree of confusion was observed between certain benign and malignant cases. This is primarily attributed to the significant visual similarities between pulmonary nodules in the early stages of the disease. Despite this, the overall performance remains exceptionally strong, positioning the model as a promising assistive tool for lung cancer diagnosis.

5.Applications

Deep learning-based lung cancer detection systems have wide applicability in modern healthcare environments. One of the primary applications is within hospital radiology departments, where CNN-based models can assist radiologists by highlighting suspicious lung nodules and prioritizing high-risk cases for further examination. This support can significantly reduce diagnostic time and minimize human error, particularly in high-volume clinical settings.

Additionally, such systems can be integrated into Computer-Aided Diagnosis (CAD) platforms to support early screening programs, especially for high-risk populations such as smokers. Early detection plays a crucial role in improving treatment outcomes and survival rates.

Furthermore, deep learning models can be utilized in telemedicine applications, enabling remote analysis of CT scans in regions with limited access to specialized medical expertise. Overall, the integration of CNN-based lung cancer detection models into clinical workflows enhances diagnostic efficiency and supports informed medical decision-making without replacing the role of healthcare professionals.

6. Ethical Consideration

The use of deep learning models in medical diagnosis raises several ethical considerations that must be carefully addressed. One of the primary concerns is patient data privacy, as medical images contain sensitive information that must be securely stored and handled. Additionally, bias within training datasets may lead to unequal diagnostic performance across different patient groups.

Furthermore, AI-based diagnostic systems should function strictly as decision-support tools rather than replacements for medical professionals. Clinical responsibility must remain with qualified physicians to ensure safe and accountable medical decisions. Addressing these ethical aspects is essential for the responsible deployment of AI systems in healthcare environments.

7. Discussion and Limitations

While the proposed model achieved high diagnostic accuracy, several limitations must be acknowledged when interpreting these results.

A primary constraint is the relatively limited size of the dataset, which consists of 1,190 images. This volume may not fully encompass the vast clinical diversity and variations found in larger populations. Furthermore, relying on a single data source might restrict the model's generalizability when applied to images captured by different scanning equipment or within diverse medical environments.

Additionally, the visual overlap between certain benign nodules and early-stage malignant tumors remains a significant challenge for automated classification models. Such similarities can lead to occasional misclassifications, a well-documented issue in the field of medical image analysis.

To address these limitations, future research should explore the integration of **Transfer Learning** utilizing pre-trained deep architectures such as **VGG16** or **ResNet**, which are capable of extracting more complex and nuanced features. Moreover, incorporating larger, multi-institutional datasets will be essential to enhance the model's reliability and its practical applicability in real-world clinical settings.

8. Conclusion

The findings of this research underscore the significant potential of Deep Learning models, particularly Convolutional Neural Networks (CNN), in automating the detection of lung cancer from Computed Tomography (CT) scans with high precision and efficiency. The proposed architecture

demonstrated exceptional performance in classifying cases into normal, benign, and malignant categories, reflecting the effectiveness of Artificial Intelligence in deciphering complex medical imagery.

The results highlight the critical importance of integrating deep learning-based diagnostic systems into clinical workflows. Such systems can substantially alleviate the cognitive burden on radiologists, mitigate human error associated with manual interpretation, and accelerate the diagnostic process—a factor of paramount importance in time-sensitive conditions like lung cancer. Furthermore, these frameworks serve as robust decision-support tools designed to augment, rather than replace, human clinical expertise.

Despite limitations regarding dataset size and the visual similarities between benign and malignant patterns, the achieved results indicate that the proposed model represents a promising step toward developing reliable, intelligent diagnostic systems. Consequently, this study recommends expanding future research by utilizing larger, more diverse datasets and implementing transfer learning techniques. Such advancements will further enhance model accuracy and reliability, facilitating their practical deployment in real-world clinical environments.

Use of Research and AI Tools

During the preparation of this research, I used Google Scholar to access and review relevant scientific papers and recent studies related to deep learning and medical image analysis. The IQ-OTH/NCCD lung cancer dataset was obtained from Kaggle and used as the primary data source for training and evaluating the proposed model.

In addition, I utilized AI-assisted tools in a supportive manner to improve language clarity, organization, and academic writing quality. These tools were used only for refinement purposes, while all research design decisions, data analysis, experimental results, and conclusions were independently developed and verified by me.

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